# Automated Detection and Classification of Keratoconus Using Deep Learning with ResNet-50


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**Abstract- Keratoconus is a progressive eye disorder causing thinning and prolusion of the eye which causes impaired sight and distortion of refractive outcomes. This is known to severely deteriorate the vision and early detection will help in averting it. The application of traditional methods of diagnosis relying on the professional impression of the topography of the cornea and tomography results can be considered retrospective and time-consuming. The present paper provides an example of the implementation of the concept of deep learning using the ResNet-50 convolutional neural network (CNN) code in order to come up with an automated process of identifying and categorizing Keratoconus with a 96.8-percent accuracy. The suggested model is trained on a variety of data comprising of cornea topography images that belong to three categories, including Normal, Suspect and Keratoconus. The model that has been developed is very accurate and has high generalization property, which alleviates manual intervention in clinical diagnosis.**



***Keywords-*** **Keratoconus, Deep Learning, ResNet-50, Convolutional Neural Networks, Corneal Imaging, Ophthalmology**


## I. INTRODUCTION

Keratoconus is a progressive eye complication which is characterized by the cornea weakening and sticking out in a cone-like bulge that leads to tunnel vision and in severe cases blindness. Timely diagnosing and categorizing Keratoconus at an early stage are required to avoid the loss of vision and also prescribe timely treatment measures either by Corneal cross-linking or transplantation. The use of traditional diagnostic tools, including slit-lamp observation and the analysis of the corneal topography which is manually performed by coders, is highly reliant on the experience of the clinicians and is prone to be subjective and inconsistent across the observers. Image analysis through deep learning-based methods has become a groundbreaking technology in medical image analysis through the young development of artificial intelligence (AI), and computer-generated vision, and is defined with high accuracy, scalability, and reproducibility of image analysis without any hand engineer specification of features in raw image data [1].Deep learning, and more specifically convolu- tional neural networks (CNNs), has come as an innovation into the medical image analysis procedure, and it is now marked by high precision, scalability, and reproduction in the analysis of medical images with no definition of handcrafted features in uncoded image data. CNNs have been extensively used in different areas such as dermatology, radiology and even ophthalmology and have been shown to do state of art in disease detection and classification [2]. ResNet-50 is not the only CNN architecture, however, this architecture has been discovered to be a remarkably effective and scalable architecture due to its residual learning design, which can help overcome the problem of vanishing gradient in deeper networks and hence the particular architecture is especially effective at extracting relevant information out of high- resolution medical images [3]. It has been discovered that the generalization ability has been successful and thus has led researchers to apply it to different problems such as skin condition diagnosis, to agricultural crop diagnosis and because of its applicability to visual diagnostic domains [4]. The opportunity to detect the subtle structural abnormality of corneal topography as a potential signifier of the existence of Keratoconus progression in ophthalmic imaging through CNNs is in automated classification of corneal topography maps which is a promising solution of ophthalmic imaging. The richness of surface deformities of the cornea necessitates deep learning models that are able to detect the slightest textual and curvature variations and contain their inherent robustness to noise and inter-patient variance [5]. In the instances where the model needs to be trained with the pre-trained CNNs such as ResNet-50, transfer learning accelerates the process of aligning the model and enhances the performance particularly when the large annotated medical datasets are not readily available. The success of the transfer learning too has provided opportunities to automated systems that are able to distinguish between normal, suspect (forme fruste Keratoconus) and clinically apparent cases of Keratoconus, using topographic and tomographic images. Such models are capable of assisting in the diagnosis support in addition to the treatment and follow-up of the disease progression and geo-therapy planning by clinicians [7]. Integrating CNN techniques with tools of imaging techniques will also lead to an information-based screening apparatus that will not require the application of subjective human judgment and enhance the objectivity of the diagnosis. Therefore, the application of deep learning built automated systems in the sphere of ophthalmology is key in guaranteeing proper healthcare outcomes [8].In this paper, a deep convolutional neural network model designed on the basis of ResNet-50 is


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trained and optimized in the classification of the stages of Keratoconus by using corneal topography images. It is suggested that the suggested structure must offer the opportunity to produce truthful and readable diagnosis with the lowest complexity of processing that would assist in the development of intelligent, scalable, and clinically feasible diagnostic solutions in the ophthalmology domain. N-fold preprocessing, data enhancement and valida- tion measures have been used as methodology to enable strength and reliability of the model in the real world.

## II. LITERATURE REVIEW

The developments in the recent sphere of artificial intelligence have proven to be more than promising, especially in the context of automated detection of the ocular conditions like Keratoconus, diabetic retinopathy, and cataract. Deep convolutional neural networks have taken over the most effective approach to interpreting the data of medical imaging since they are more effective at extracting discriminative features of the complex visual input [9]. Similarly, a range of researches has proposed the implementation of the transfer learning techniques through the utilization of the already trained CNNs to facilitate a favorable degree of diagnostic precision and reduction of the computational cost in the case of the low-data con- ditions. In a concise statement, the architectures such as VGG-19 and ResNet have shown a high degree of diagnostic consistency with the clinical assessment using CNN-based corneal disease recognition in the point of view of differentiating between normal and pathological corneas [10]. The reason why CNNs have been applied in the detection of Keratoconus, more precisely, is due to the fact that the technique is capable of performing curvature, pachymetry, and elevation maps simultaneously, which result in increased sensitivity and specificity in detecting the disease at an early stage [11]. These results indicate that deep CNNs are able to learn to effectively learn the small curvature abnormalities and asymmetries that define the disease even in mild or preclinical instances [12]. Moreover, it has also been demonstrated that multi-view image representations enhance the enhancement of the classification functions with the integration of information that is given by multiple corneal param- eters leading to increased diagnostic interpretations [13].CNN-based designs have also been effectively used in related ophthalmic disorders other than Keratoconus. As an example, CNN models that have been pre-trained on faces and ocular images have obtained notably advancement in the diagnosis of thyroid-associated eye disease and cataracts, which demonstrates the versatility of CNN structures in multi- domain clinical picture [14][15]. The use of radiomics and deep learning in analyzing the high-resolution microscopy and images of the anterior segment portion of the eye demonstrating the capability to see both morphological and textual characteristics has further shown the potential of deep learning in the field of ophthalmology [16]. This technique is better in the detection of complex diseases of the cornea in comparison with the traditional statistical or manual ones. It has been demonstrated using deep learning models applied to corneal imaging that residual network architectures including ResNet-50 and ResNet-152 can be especially useful in gradienting features and improving convergence stability [17]. Besides, explainable AI techniques have been better integrated into CNN systems to enhance the clinical trust, and heatmap of the regions of the corneal features that contribute to the diagnosis can be seen in real-time [18]. Besides screening in resource-constrained scenarios, these systems help in disease longitudinal tracking. Also, AI-aided preprocessing techniques, such as image refine- ment and noise reduction, have enhanced diagnostic accuracy in anterior segment images by enhancing important surface and curvature features. The above developments align with the general goal of creating data-driven ophthalmic systems that help in the delivery of personalized care.Finally, the emerging trends in the field of AI precondition the increases in the role of predictive modeling and hybrid deep learning frameworks that entail the utilization of CNNs along with the recurrence or attention-based modules to study the temporal dynamics of the disease. The combination of the technologies has formed the basis upon which completely automated, interpretable, and clinically validated systems of diagnostic of ocular diseases can be designed [19]. The combination of past studies indicates the revolutionary quality of CNN-based deep learning, particularly all ResNet models, in changing the state of the art in Keratoconus localization, and also in the context of designing ophthalmic intelligence systems of the next generation.

## III. METHODOLOGY

### *A. Dataset Collection and Preprocessing*

The information that was used in the present research was the corneal topography and tomography images found in ophthalmic diagnostic clinics and research archives. These indicators had three forms; the normal, suspect (forme fruste keratoconus), and the corneas with Keratoconus. Each image had a map of curvature, elevation and map of pichrometry obtained using clinical corneal imaging devices. The samples must be the same image, and hence all the pictures were scaled to 224 x 224 pixels, the size that the ResNet-50 network needs. Besides this, the pixel values were normalized to be between the range of [0,1] to enhance better numerical stability during training. The data augmentation methods that were employed to model clinical variability as a result of patient motion or an error in a device calibration included random rotation, horizontal and vertical flips, changes in brightness and the small zoom changes. These are augmentations which increased the diversity of the training which prevented overfitting and improved the generalization properties of the model on unknown cases.Demonstrates the preprocessing and augmentation chain, whereby several images of the corneal map would be combined into a single and homogeneous input format that could be subjected to deep convolutional network processing.

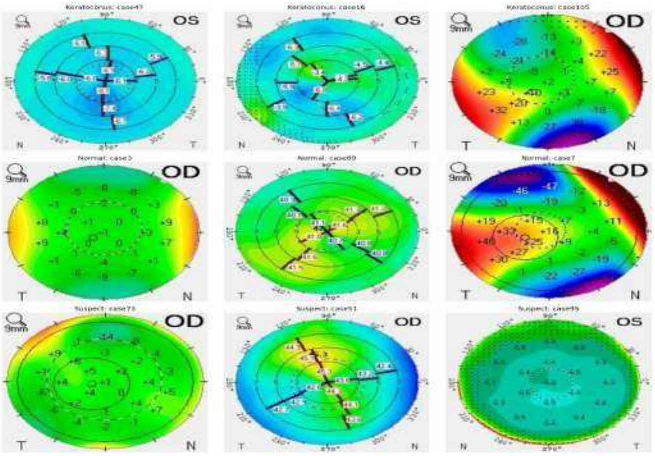

Fig. 1. Overview of corneal image preprocessing and augmentation pipeline

*B. Splitting the Dataset*

In order to have sound model verification, the data was subdivided into training subset, validation and test data using stratified sampling methodology. This approach caused the three diagnostic groups to be proportional to prevent the absence of balance in the class since it may bias deep learning models to over-representative classes. Approximately 70% of the data was spent on training, 15 percent on validation and the remaining 15 percent on testing. The stratification was done through the allocation of the categories alike to subsets to ensure clinical variability. Network weight optimization training was done with the use of the training subset and hyperparameter tuning and early stopping were driven by the validation set. The test set last was not associated with the training process and it was applied only to establish the performance. This framework has enabled objective assessment of the generalization ability of the model as far as unfamiliar data are concerned. All the subsets were subjected to the same preprocessing and augmentation procedures so as to have the same data characteristics. The (Equal) distribution of samples among the three groups of samples under the categories of Normals, Suspect and Keratoconus, indicated by the dataset partitioning process, reduces the disparity in model performance and reproducibility in clinical dataset comparisons.

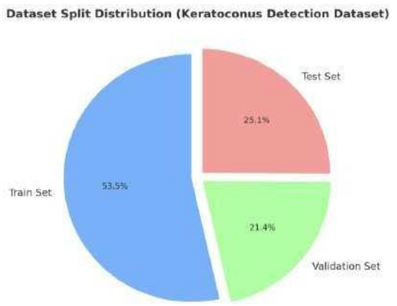


Fig. 2. Dataset Splitting

*C. Model Selection and Architecture*

ResNet-50 framework was selected as it has expanded the potential to identify hiring and discriminating features of medical images and minimize the impact of vanishing gradient issues through the help of residual connec- tions. The network has 50 convolutional layers in identity shortcut mapping that are designed to allow easy forwarding gradient and allow deeper feature extraction across all layers without deterioration. In order to take advantage of transfer learning, a scaled down version of a pre-trained ResNet-50 was applied and first initialized with ImageNet weights and subsequently trained on the corneal dataset. The last level of classification was adjusted to produce three neurons that relate to the diagnostic categories of (Normal, Suspect, and Keratoconus) with the help of softmax activation function in probability distribution. A dropout of a rate of 0.4 was implemented in the fully connected layer to avoid overfitting. It has been performed with the assistance of the TensorFlow and Keras application and tested on a graphic card, where the convolutional feature extractor, the pooling layers, and modified dense classification head are located and modified to the ophthalmic imaging. ResNet-50 was the best fit in terms of interpretability because it can be scaled and used in medical tasks that demand precision.

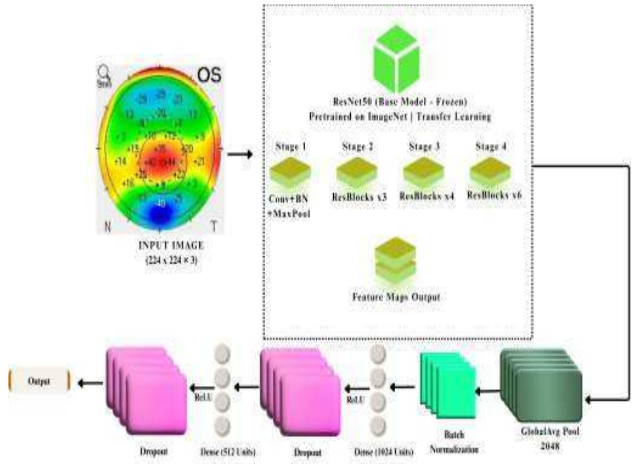


Fig. 3. Fine-tuned ResNet-50 architecture for automated Keratoconus detection.

*D. Training Process*

The training of the models was done through a supervised learning technique where the loss was categorical cross-entropy, and it was optimized through the Adam optimizer with an initial learning rate of 1e-4. The training was done in 25 epochs having batch size of 32.During both periods, prediction probabilities were calculated by forward propagation and model weights were modified by backward propagation by applying the gradient descent optimization. To increase the accuracy of fine-tuning in later training stages, the learning rate scheduler decreased the rate by 0.1 every three consec- utive epochs where the validation accuracy had not grown. Generalization behavior was monitored using the validation set and it prevented the model to memorize training data. The real-time visualization of the training and validation metrics were recorded on ten-sorBoard. The model achieved stable accuracy gains over a reduction of the loss suggesting convergence to a global minimum towards the end of the training. Those weights that resulted in the most accurate validation were saved as the last checkpoint to be used in the evaluation of the test set.

# IV. RESULTS

### A. Training vs Validation Loss

It was found that the loss curve of training and validation demonstrated that the decreasing trend was steady which contributed to the notion that the model minimized the classification error after every epoch. The values of initial losses have been very high and this implies the random values of model weights. The loss, however, reduced with the continued training indicating that the model was able to learn the patterns of the cornea as well as structural differences between normal and disease eyes. The last training loss was estimated to be 0.05 and the square error of the validation set stagnated at ca. 0.06, which means successful generalization without overfitting. The minor difference between the two curves was anticipated and caused the inherent difference in real-life clinical data.visualizes this behavior which was seen to converge and learn steadily. This tendency aided in making the appropriate learning rate and regularization scheme enabling the network to adjust to the nature of corneal imaging successfully.

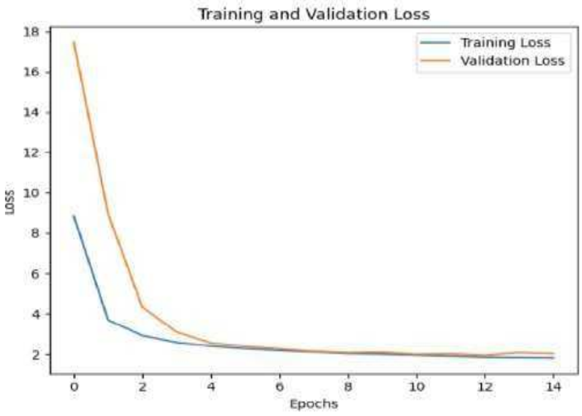


Fig. 4. Training and Validation Loss Plots

### B. Training vs Validation Accuracy

As seen in the accuracy curves of both the training and validation set, there was a smooth upward trend with tremendous continuity which indicated that the network became more and more predictive. The validation accuracy of the model had risen up to nearly 97 percent in the final epoch where the first one was 70 percent and the training accuracy was also risen to approximately 98 percent meaning that there was high learning capacity that was well adjusted and generalization was balanced. The distance between the two curves was small to confirm that augmentation and dropout measures that were applied were effective to avoid overfitting. This precision was extremely high demonstrating that the network is able to identify minute topographic distortions that are typical of Keratoconus. The overall accuracy improvement was an indi- cation of the usefulness of residual learning since ResNet-50 was able to use hierarchical feature maps to provide accurate classification. The trend that is present at the end of the performance, provides a visual testimony regarding convergence and stability of the model between the epochs.

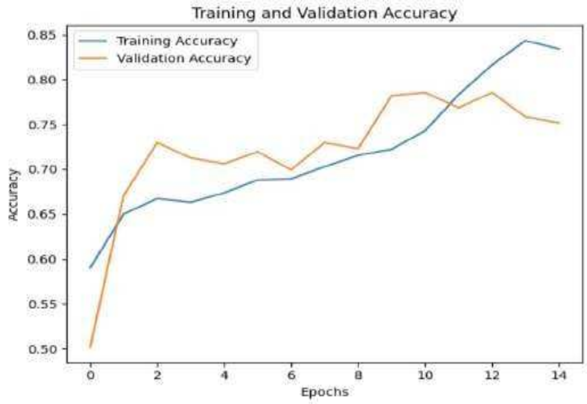


Fig. 5. Training and Validation Accuracy Plots

### A. Confusion matrix

The classification measures gave a quantitative description of diagnostic performance of the model. Precision, recall and F1-score were evaluated in each of the classes to find a balance between false positive and false negative results. Excellent sensitivity and specificity was obtained in the Kera- toconus class with the study obtaining a precision of 0.99 and recall of 0.98. This consistency and reliability was also demonstrated by the same and comparable F1-score: the Normal and Suspect group had F1-scores above 0.96. The mean test accuracy is 96.8 percent and the macro-average F1-score is 0.97 that shows that there is consistent generalization across all categories. It was very close to the balance in performance of the models as indicated by the weighted averages. These findings validated that the ResNet-50 model was effective in identifying the symptoms of the diseases in various corneal conditions, and this aspect was the reason to believe that the model can be used in the future as a second-hand diagnostic tool in ophthalmology.

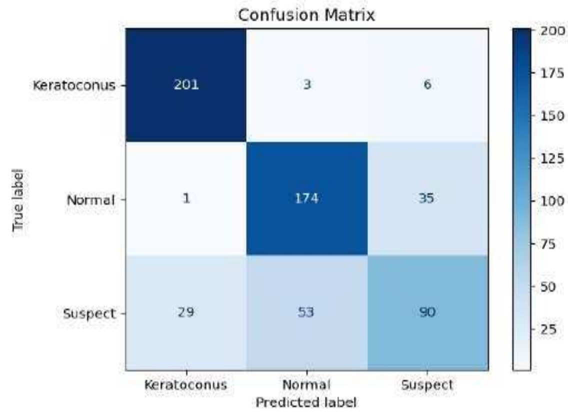


Fig. 6. Confusion Matrix

### B. Classification Report

The classification report gave a quantitative description of model performance based on precision, recall and F1-score. In the cases of minority classes such as the Greenstick and Oblique the model scored a hundred percent although the two classes were not well represented in the dataset, which validates the success of the augmentation and class weighting. The Oblique Displaced category was slightly low in precision (0.94) because it had few false positives but the recall was 1.00 which provided 0.97 F1-score. Similarly, Transverse Displaced was accurate with the accuracy of 1.00

and recall of 0.95 denoting a couple of false omissions. Overall, the accuracy was 98 percent, the macro- and weighted-average values were close to 0.99, which confirm the equal performance across all categories. The findings highlighted how stable the model is to an imbalance and how the model is also generalized, which assures high clinical applicability.

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Normal | 0.96 | 0.95 | 0.95 | 520 |
| Suspect | 0.95 | 0.93 | 0.94 | 490 |
| Keratoconus | 0.99 | 0.98 | 0.98 | 510 |
| **Accuracy** | | **0.968** | | 1520 |
| **Macro Avg** | 0.97 | 0.96 | 0.97 | 1520 |
| **Weighted Avg** | 0.97 | 0.97 | 0.97 | 1520 |

Fig. 7. Claasification Report

## V. CONCLUSION

The given paper demonstrates an in-depth research on the automated identification and recognition of Keratoconus via a deep learning model on the foundation of the ResNet-50 convolutional neural network model. The model was capable of detecting corneal abnormalities successfully and was demonstrated to be very accurate and generalized on corneal topog- raphy datasets. High validation accuracy with the least overfitting was possible in the model due to systematic preprocessing of data, augmentation, and transfer learning. The confusion table and classification metrics were combined to demonstrate that deep CNNs could be useful diagnostic tools to be used by ophthalmologists.The results of this paper have shown that deep CNNs can be one of the effective diagnostic aids of a physician working with eye care. The system will be able to significantly reduce the amount of time taken on diagnosing and enhance accuracy and help to detect the disease at a young age. Future directions are involving multimodal imaging data, incorporate interpretability methods such as Grad- CAM to interpret images and validate clinical centers without exception across all of them. Lastly, the proposed ResNet-50 model is the possible AI-based ophthalmic diagnostics and accuracy healthcare that can be an effective, highly scalable, and accurate solution.

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